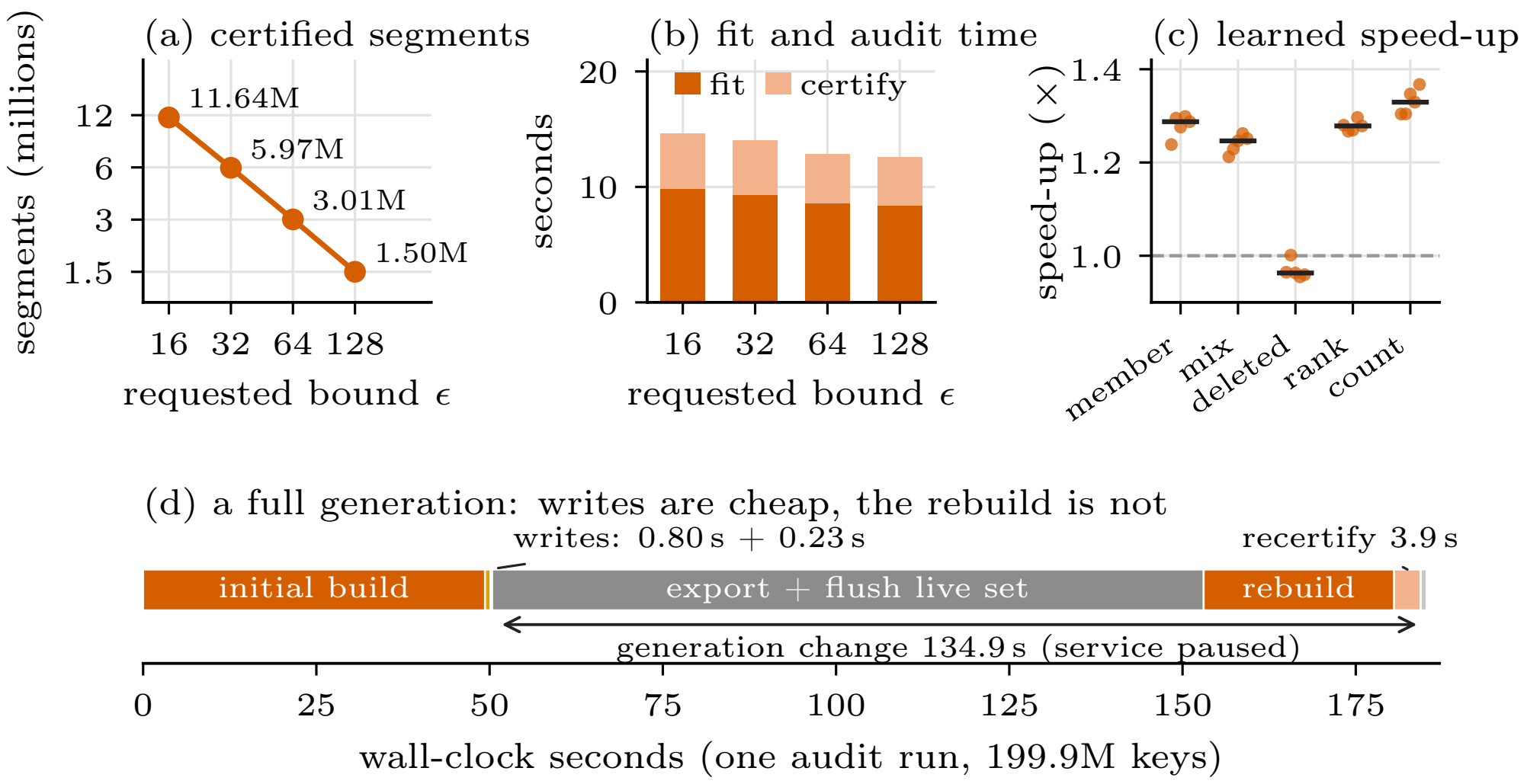

(a) certified segments
segments (millions)
12
6
3
1.5
11.64M
5.97M
3.01M
1.50M
16 32 64 128
requested bound ϵ
(b) fit and audit time
seconds
20
10
0
fit
certify
16 32 64 128
requested bound ϵ
(c) learned speed-up
speed-up (×)
1.4
1.2
1.0
member
mix
deleted
rank
count
(d) a full generation: writes are cheap, the rebuild is not
writes: 0.80 s + 0.23 s
recertify 3.9 s
initial build
export + flush live set
rebuild
generation change 134.9 s (service paused)
0 25 50 75 100 125 150 175
wall-clock seconds (one audit run, 199.9M keys)